

MySQL Assignment 2 – Querying Data

1. Distinct Values:

- a query to retrieve distinct salaries from the Employees table.

57 • `select distinct(salary) from employees;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	salary
▶	60000.00
	80000.00
	45000.00
	70000.00
	55000.00
	65000.00
	90000.00
	95000.00
	75000.00
	85000.00
	100000.00

2. Alias (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

59 • `select age as Employee_Age, salary as Employee_Salary from employees`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Employee_Age	Employee_Salary
▶	27	60000.00
	31	80000.00
	26	45000.00
	29	70000.00
	25	55000.00
	28	65000.00
	32	90000.00
	27	70000.00
	30	55000.00
	29	80000.00
	28	60000.00
	26	95000.00

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3. Where Clause & Operators:

© Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
1  select * from employees where salary > 50000 and hire_date < '2016-01-01'
```

[illegible]

🔴 Find the employee whose designation is missing and fill it with "Data Scientist".

```
2 select * from employees where designation is NULL
3
4 update employees set designation = "Data Scientist" where employee_id = 5004
5
```

[illegible]

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Sorting and Grouping Data:

1. ORDER BY:

- 🔴 Find employees sorted by department ID in ascending order and salary in descending order.

```
6 select * from employees order by department_id asc,salary desc
```

employee_id	employee_name	gender	age	hire_date	designation	salary	department_id
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	2
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	75000.00	3

2. LIMIT:

- **Display the first 5 employees hired in the year 2018.**

```
8 select * from employees where hire_date between '2018-01-01' and '2018-12-31' limit 5
```

[illegible]

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3. Aggregate Functions:

- ③ Calculate the sum of all salaries in the Finance department.

```
13 select sum(salary) from employees
14 where department_id = (select department_id from department where department_name = 'finance')
15
16 select sum(e.salary) as total_salary from employees e
17 inner join department d on e.department_id = d.department_id
18 where d.department_name = "finance"
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
total_salary				
▶	170000.00			

- ③ Find the minimum age among all employees.

```
20 select min(age) from employees
21
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
min(age)				
▶	25			

4. GROUP BY:

- ③ List the maximum salary for each location.

```
20
21 select l.location,max(e.salary) from employees e
22 inner join location l on e.location_id = l.location_id group by l.location
23
24
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
location max(e.salary)				
▶	Bangalore 95000.00			
	Chennai 95000.00			
	Hyderabad 90000.00			
	Pune 100000.00			

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- ② Calculate the average salary for each designation containing the word 'Analyst'.

```
20 select designation,avg(salary) from employees
21 group by designation having designation like'%Analyst%'
```

designation	avg(salary)
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

- ② Find departments with less than 3 employees.

```
23 select d.department_name, count(e.employee_id) from employees e
24 join department d on e.department_id = d.department_id group by d.department_name
25 having count(e.employee_id) < 3
26
27
```

department_name	count(e.employee_id)
Business Development	2
Content Creation	2
Finance	2
Operations	1
Product Management	2
Research and Development	1
Software Development	1

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- ② Find locations with female employees whose average age is below 30.

```
30 select l.location, avg(e.age) from employees e
31 join location l on e.location_id = l.location_id where e.gender = "female"
32 group by l.location having avg(e.age) < 30
33
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
location	avg(e.age)							

Joins:

1. Inner Join:

- ② List employee names, their designations, and department names where employees are assigned to a department.

```
31 select e.employee_name, e.designation, d.department_name
32 from employees e
33 inner join department d
34 on e.department_id = d.department_id;
35
36
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	employee_name	designation	department_name					
▶	Anvi Chaudhary	Business Development Executive	Business Development					
	Kiaan Desai	Sales Executive	Business Development					
	Ananya Paul	Content Writer	Content Creation					
	Anaya Kapoor	Event Coordinator	Content Creation					
	Aaradhya Iyer	Customer Support Executive	Customer Support					
	Arnav Rao	Customer Success Manager	Customer Support					
	Vihaan Mohan	Supply Chain Analyst	Customer Support					
	Vihaan Singh	Data Analyst	Data Science					

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2. Left Join:

☉ List all departments along with the total number of employees in each department, including departments with no employees.

```
36 • select d.department_name, COUNT(e.employee_id) AS total_employees
37 from department d
38 LEFT JOIN employees e
39 ON d.department_id = e.department_id
40 GROUP BY d.department_name;
41
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
employee_name	designation	department_name	
Anvi Chaudhary	Business Development Executive	Business Development	
Kaani Desai	Sales Executive	Business Development	
Ananya Paul	Content Writer	Content Creation	
Anaya Kapoor	Event Coordinator	Content Creation	
Aaradhya Iyer	Customer Support Executive	Customer Support	
Arnav Rao	Customer Success Manager	Customer Support	
Vihaan Mohan	Supply Chain Analyst	Customer Support	
Vihaan Singh	Data Analyst	Data Science	

3. Right Join:

☉ Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
42 • select l.location, e.employee_name
43 from location l
44 left join employees e
45 on l.location_id = e.location_id;
46
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
location	employee_name		
Bangalore	Dhruv Shetty		
Bangalore	Myra Verma		
Bangalore	Vihaan Mohan		
Bangalore	Kabir Nair		
Bangalore	Anika Paul		
Bangalore	Vivaan Singh		
Bangalore	Arjun Kumar		
Chennai	Reyansh Singh		